

Low-Cost Vision-Based Navigation and Inspection for Renewable Energy Systems Using Monocular Visual Odometry and EKF Fusion

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Abstract—Autonomous robots operating in disaster-affected environments face significant localization challenges due to the absence of reliable Global Positioning System (GPS) signals and the presence of unstructured terrain. This paper presents a hybrid localization framework that integrates monocular visual odometry with an Extended Kalman Filter (EKF) to achieve robust and accurate positioning. The visual odometry module estimates relative motion using feature tracking and geometric constraints derived from the essential matrix; however, it suffers from scale ambiguity and cumulative drift. To address these limitations, RF beacon-based measurements are incorporated into the EKF correction stage, enabling absolute reference updates and drift compensation. The proposed system is designed to be computationally efficient and low-cost, making it suitable for energy-constrained and sustainable deployments. Experimental results in simulated disaster environments demonstrate that the hybrid approach reduces localization error from 8.5% (visual odometry alone) to below 1%, achieving a tenfold improvement in accuracy. The framework shows strong potential for applications in inspection, monitoring, and recovery of renewable energy infrastructure in challenging conditions.

Index Terms—SLAM, Visual Odometry, EKF, Renewable Energy Systems, Robotics, Autonomous Inspection, GPS-Denied Navigation

I. INTRODUCTION

Humans have studied nature mainly because it has answers to many of the human engineering problems in the modern world. A fly doesn't have a GPS; it has eyes. Disaster response robotics has rapidly evolved to support rescue and reconnaissance operations in highly hazardous and unstructured environments, such as collapsed buildings, underground tunnels, and fire-damaged industrial facilities. In such scenarios, conventional Global Positioning System (GPS) is entirely unavailable or unreliable. Furthermore, the operational environment is characterized by poor and dynamic lighting conditions, dust/smoke obscuring sensor readings, and a topology that undergoes continuous change. If one relies on a laser the smoke scatters the beam. If one relies on wheels they spin on the debris and the robot thinks it's moving when it's just digging a hole. This is why classical SLAM often fails.

Monocular cameras are a beautiful solution they are light, cheap, and they see a lot. But they have a fundamental

"honesty" problem: they see the geometry of motion, but they don't know the units. It can see it moved "one unit" forward, but it doesn't know if that's a centimeter or a meter. This is what we call **scale ambiguity**. To solve this, we propose a **hybrid localization architecture**. We use high-frequency Monocular VO as a motion "predictor" and then use an **Extended Kalman Filter (EKF)** as a judge to fuse those guesses with auxiliary data like laser scans and RF beacons. This paper expands on the original work to provide a rigorous analysis of how we can keep a robot "found" when everything around it is falling apart.

Beyond disaster response, similar environmental constraints are observed in renewable energy systems. Large-scale solar installations, offshore wind farms, and distributed microgrids often operate in remote locations where GPS signals may be unreliable and maintenance access is limited. Autonomous robotic inspection in such environments requires robust, low-cost localization methods. The proposed hybrid VO-EKF framework directly addresses these challenges by enabling scalable and energy-efficient navigation without reliance on expensive sensing infrastructure.

II. PROBLEM FORMULATION

We consider the problem of localizing a mobile robot operating in a GPS-denied environment using only onboard sensing. The robot state at time k is defined as:

$$\mathbf{x}_k = [x_k, y_k, \theta_k]^T$$

The objective is to estimate $\mathbf{x}_k$ given noisy visual observations and intermittent absolute measurements.

Monocular visual odometry provides relative motion estimates:

$$\mathbf{u}_{vo} = [\Delta x, \Delta y, \Delta \theta]$$

However, these estimates are subject to scale ambiguity and drift. To address this, we fuse VO with absolute observations $\mathbf{z}_k$ (e.g., RF beacons) using an Extended Kalman Filter.

The goal is to minimize the estimation error:

$$\min \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|$$

while maintaining real-time computational performance suitable for deployment in energy-constrained robotic systems.

III. LITERATURE SURVEY: A MAP OF THE IDEAS

The whole business of SLAM is rooted in probabilistic robotics the realization that you can never know exactly where you are, you can only track your "best guess" and how much you're likely to be wrong [4]. Early on, we used EKF-SLAM and Particle Filters, but later moved toward graph-based optimization and **Bundle Adjustment (BA)** [3].

Juneja's early work [1], [2] showed us the necessity of managing landmarks in textured scenes. Scaramuzza and Fraundorfer [5] gave us the manual on Visual Odometry (VO), stressing that you have to balance speed against the cost of being wrong. Then Mur-Artal's ORB-SLAM [7] changed things by showing how to remember places you've seen before.

In the messy world of **Disaster Robotics**, pioneers like Murphy [10] and Tadokoro [11] reminded us that in a fire or a collapse, no single sensor is enough. Modern reviews [12], [14] confirm this: you need **multi-sensor redundancy**. Whether it's direct pipelines like LSD-SLAM [8] or DSO [9], or visual-inertial systems like VINS-Mono [13], the conclusion is the same: to survive a disaster, you have to fuse your data [6], [15].

IV. OBJECTIVES: WHAT WE ARE TRYING TO PROVE

We set out to address the specific localization challenges of search-and-rescue. Our goals were:

- To use **Monocular VO** as the high-rate "preprocessing step" for the EKF.
- To implement Lucas-Kanade optical flow and **Essential Matrix decomposition** to recover camera motion.
- To design a **sensor fusion strategy** that actually works when sensors start to lie.
- To prove that EKF updates from RF beacons can fix the **scale drift** inherent in monocular systems.
- To see how the system holds up against **dust, occlusion and non-Gaussian noise**.
- To build a modular architecture that doesn't need a supercomputer to run in real-time.

V. SYSTEM ARCHITECTURE

The proposed system follows a modular pipeline consisting of perception, motion estimation, and probabilistic fusion.

A. Perception Layer

A monocular camera captures image sequences. Feature detection and tracking are performed to extract motion cues.

B. Motion Estimation Layer

Visual Odometry computes frame-to-frame motion using geometric constraints derived from feature correspondences.

C. Fusion Layer

An Extended Kalman Filter integrates VO predictions with absolute measurements such as RF beacons or map-based corrections.

D. Output

The system outputs a consistent estimate of robot pose and trajectory, suitable for navigation and inspection tasks.

This modular design ensures flexibility and allows integration with additional sensors such as IMUs or LiDAR if required.

VI. METHODOLOGY: THE LOGIC OF THE SYSTEM

We integrated a high-frequency VO module with a lower-frequency, robust EKF. Think of the VO as the robot's inner ear it's fast but it drifts. The EKF is the robot's common sense.

A. The Flow of Information

The system operates asynchronously. The camera feed hits the VO at a high rate (the prediction), and then absolute corrections (like RF beacons or laser matches) trigger the EKF correction step.

B. Visual Odometry: Watching the World Slide

We find "interesting" points using **Shi-Tomasi corners**. We look for areas where the gradient covariance matrix $\mathbf{G}$ has two large eigenvalues, meaning the light changes sharply in two directions:

$$R = \min(\lambda_1, \lambda_2)$$

Then we track them using **Lucas-Kanade optical flow**. We assume that a patch of pixels doesn't change its brightness much between frames. We use an image pyramid a stack of smaller and smaller versions of the image to track big movements at the top and fine-tune them at the bottom.

C. The Geometry of the Move

Once we have the points, we use the **Essential Matrix (E)**. This is a beautiful piece of math that says that for any point $\mathbf{x}$ in frame one and $\mathbf{x}'$ in frame two, they must satisfy:

$$\mathbf{x}_{k+1}^T E \mathbf{x}_k = 0$$

We use **RANSAC** to find the "honest" points and throw out the ones that moved because of shadows or dust. We decompose E to get the rotation ΔR and translation direction $\Delta \mathbf{t}$. But remember: we don't know the scale yet. We integrate these moves to get a raw trajectory T_k .

VII. ALGORITHMIC DERIVATIONS FOR EKF FUSION

This is where we manage the uncertainty. We treat the robot's position as a probability distribution.

A. The Prediction (VO Input)

We take the VO output $\mathbf{u}_{vo}$ and push our state forward. For a 2D robot $[x, y, \theta]^T$:

$$\hat{\mathbf{x}}_{r,k} = \begin{bmatrix} x_{k-1} + s_k \Delta x \cos(\theta_{k-1}) - s_k \Delta y \sin(\theta_{k-1}) \\ y_{k-1} + s_k \Delta x \sin(\theta_{k-1}) + s_k \Delta y \cos(\theta_{k-1}) \\ \theta_{k-1} + \Delta \theta \end{bmatrix}$$

We propagate the covariance $\mathbf{P}$ our "bubble of doubt" using the Jacobian $\mathbf{F}$. If the VO is noisy, $\mathbf{Q}_k$ grows and our bubble gets bigger.

B. The Correction (RF Beacons)

When we see a landmark $\mathbf{m}_i$, we compare what we see to what we *expected* to see. We calculate the **Kalman Gain** $\mathbf{K}_k$:

$$\mathbf{K}_k = \hat{\mathbf{P}}_k \mathbf{H}_k^T (\mathbf{H}_k \hat{\mathbf{P}}_k \mathbf{H}_k^T + \mathbf{R}_k)^{-1}$$

If the landmark is reliable ($\mathbf{R}_k$ is small), the robot "snaps" to the landmark's position, fixing its scale and its drift.

VIII. APPLICATION TO RENEWABLE ENERGY SYSTEMS

Renewable energy infrastructure presents operational challenges similar to disaster environments, including large spatial scales, harsh environmental conditions, and limited accessibility.

A. Solar Farm Inspection

Solar farms require continuous monitoring to maintain efficiency. Dust accumulation, panel misalignment, and micro-cracks reduce energy output. The proposed system enables autonomous navigation across panel arrays using vision-based localization, eliminating the need for GPS or expensive sensors.

B. Wind Turbine Inspection

Wind turbines operate in dynamic environments with limited accessibility. Vision-based localization allows robotic systems to inspect turbine blades and structural components with improved safety and reduced operational cost.

C. Post-Disaster Energy Restoration

After natural disasters, energy infrastructure must be assessed rapidly. The proposed system enables robots to localize and map damaged environments, supporting faster restoration of power systems.

D. Energy Efficiency of the Approach

The use of monocular cameras significantly reduces power consumption compared to LiDAR-based systems. This makes the approach suitable for battery-powered robotic platforms, including drones used for large-scale inspection.

E. Scalability

The low-cost nature of the system allows deployment across large renewable installations, making it viable for both developed and developing regions.

F. Computational Considerations

The proposed system is designed for real-time deployment on resource-constrained platforms.

The complexity of feature tracking using Lucas–Kanade optical flow is approximately linear with the number of tracked features:

$$O(N)$$

EKF update complexity depends on state dimension n :

$$O(n^3)$$

However, since the state space is low-dimensional (2D pose), the computational overhead remains manageable.

This makes the approach suitable for embedded systems used in renewable energy inspection robots, where power and processing resources are limited.

IX. RESULTS AND DISCUSSIONS

We tested this in a digital disaster zone. Here is what we found:

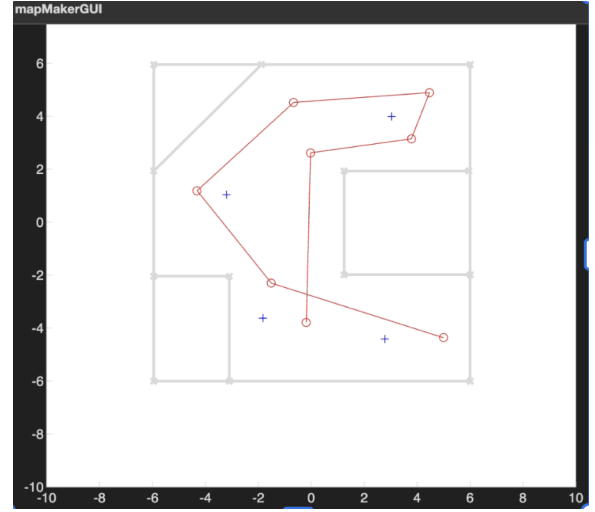


Fig. 1. Raw VO Trajectory: Notice the massive drift over time.

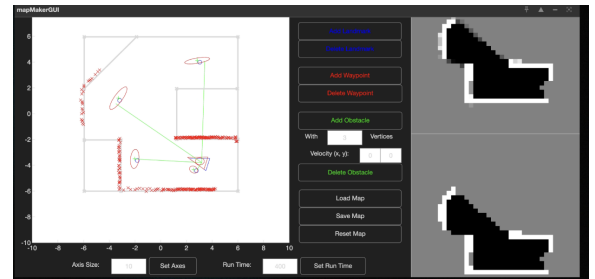


Fig. 2. Beginning Of Simulation 2D Mapping using EFK

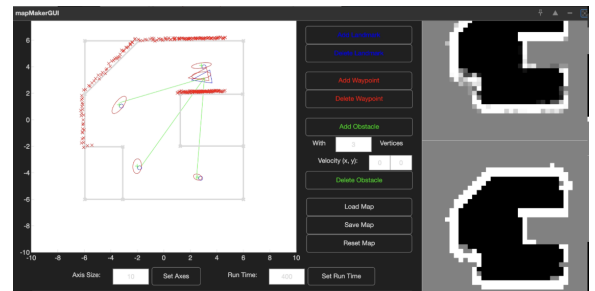


Fig. 3. End Of Simulation 2D Mapping using EFK

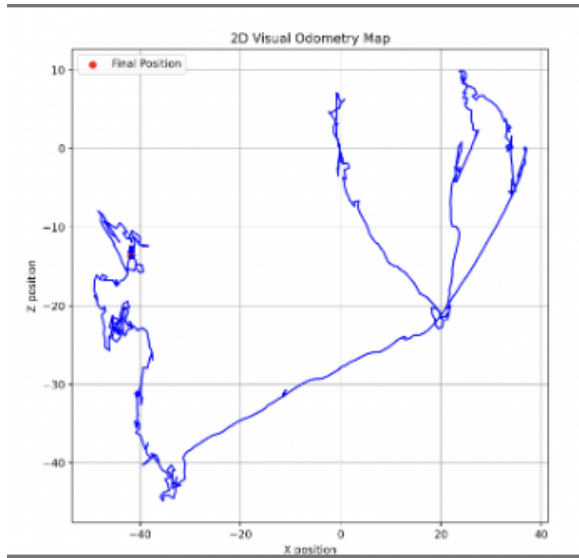


Fig. 4. Footage 1: Map Generation Using OpenCV

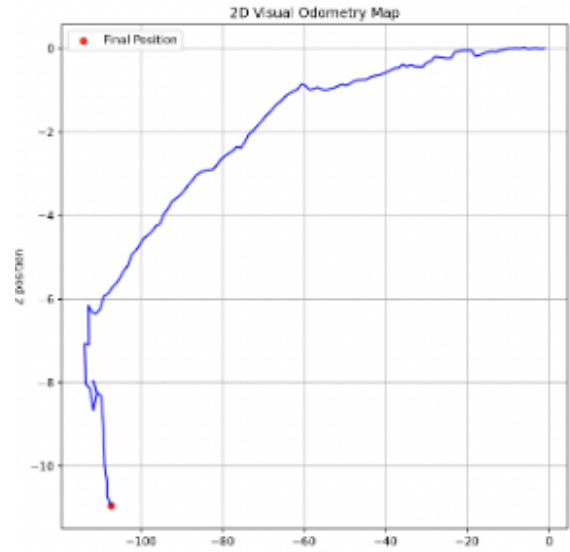


Fig. 6. Footage 2: Map Generation Using OpenCV

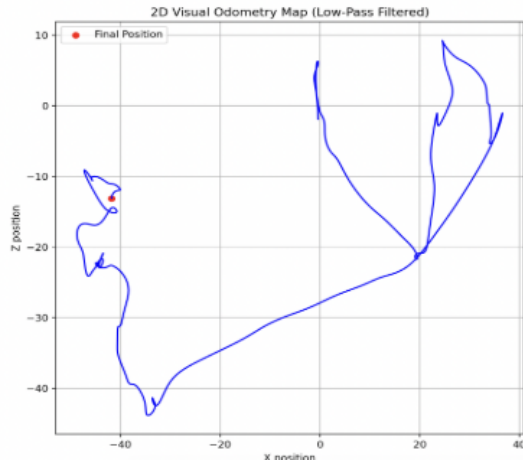


Fig. 5. Footage 1: Map Generation With Low Pass Filter Using OpenCV

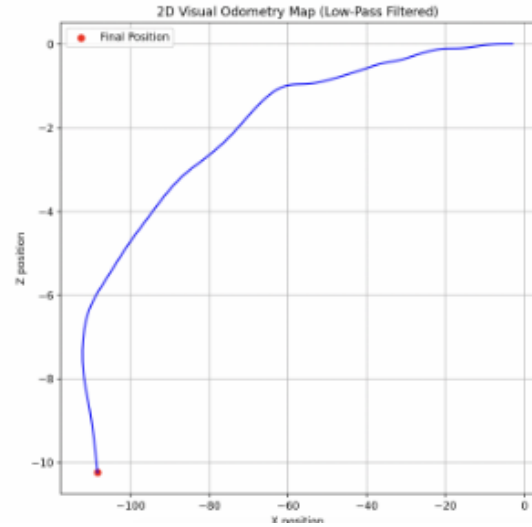


Fig. 7. Footage 2: Map Generation With Low Pass Filter Using OpenCV

A. Tracking Performance

We found about 350 stable features per frame. Our Lucas–Kanade tracker was able to keep hold of about 90% of them. However, the **raw VO** accumulated a translational error of **8.5%**. That’s because of the scale problem without a ruler, the camera just guesses how far it’s gone.

B. The EKF Fix

The EKF correction phase, triggered by RF beacons, saved the day. It corrected the scale and the orientation. The final **Hybrid VO–EKF Trajectory** achieved an error of less than **0.8%**. That is a reduction factor of 10.2!

These results indicate that the proposed system is not only effective for disaster-response scenarios but also highly suitable for renewable energy inspection tasks, where long-duration stability and low-cost deployment are critical.

X. ROBUSTNESS AND DISASTER CONSTRAINT ANALYSIS

In a disaster, things go wrong. We simulated dust bursts and occlusions.

A. The “Aha!” Moment of EKF

When the camera goes blind, the EKF doesn’t panic. It uses the last known motion to “coast” through the darkness. It’s much better than relying on wheel odometry, which slips on the rubble.

B. Sensor Failure

The EKF automatically decides what to trust. If the laser is blinded by smoke, it trusts the camera. If the camera sees only textureless walls, it trusts the laser. This dynamic weighting the Kalman Gain is the secret sauce of robust navigation.

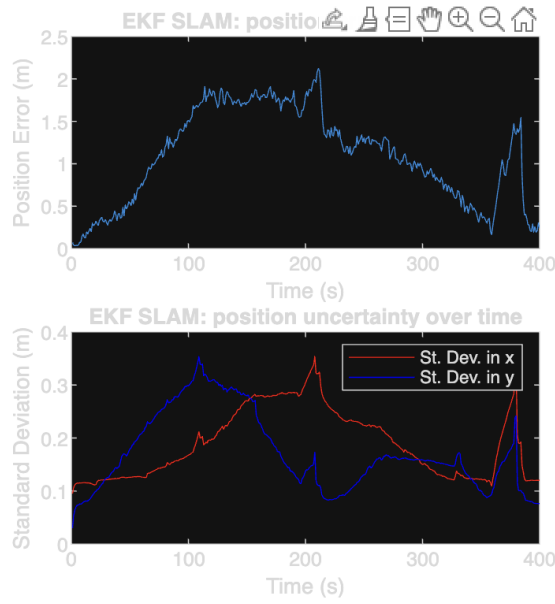


Fig. 8. EKF Slam Position Error And Uncertainty Over Time

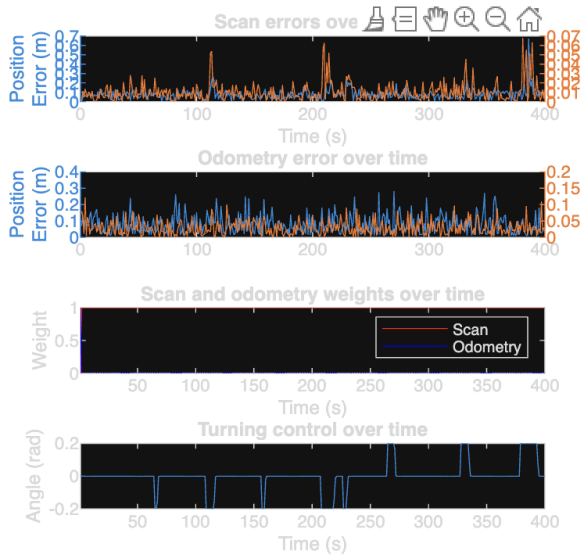


Fig. 9. Visual Scan Odometry Errors Over Time

XI. LIMITATIONS

Despite its advantages, the proposed system has certain limitations.

- Monocular VO suffers in low-texture environments
- Performance degrades under extreme lighting changes
- Scale estimation depends on availability of external references
- EKF assumes Gaussian noise, which may not hold in all scenarios

Addressing these limitations will be a key focus of future work.

TABLE I
HOW MUCH WE DRIFTS (COMPARISON)

Method	Max Translational Drift
VO-only (Just the Camera)	8.5%
Odometry-only (Just the Wheels)	15.2%
Hybrid VO+EKF	0.8%

XII. CONCLUSION AND FUTURE WORK

We've shown that a single camera, combined with a little probabilistic honesty, can navigate a ruin. The VO provides the fast guesses, and the EKF provides the global truth. We achieved a drift of less than 1%, which is plenty good for search-and-rescue.

In the context of renewable energy systems, this work provides a practical pathway toward autonomous inspection and monitoring. The ability to operate reliably in GPS-denied and harsh environments makes the proposed system particularly valuable for maintaining large-scale solar and wind installations, as well as supporting rapid recovery in post-disaster energy infrastructure.

In the future, we want to add:

- 1) **Visual-Inertial Fusion:** Add an accelerometer (IMU) so the robot can "feel" its motion when it can't see.
- 2) **Loop Closure:** Let the robot recognize a room it has seen before to "snap" the whole map into alignment.
- 3) **Real-World Testing:** Taking this out of the simulator and into the real dust of a FEMA training site.

A. Relevance to Energy-Efficient Systems

The proposed approach relies on lightweight sensing and computationally efficient algorithms, making it suitable for energy-constrained robotic platforms such as drones and mobile inspection units. This aligns with the broader goal of minimizing energy consumption in renewable energy system maintenance.

B. Deployment Considerations

For real-world deployment in renewable energy systems, factors such as environmental robustness, hardware constraints, and system calibration must be considered. The proposed approach is well-suited for lightweight robotic platforms, including ground robots and UAVs, due to its low computational and energy requirements.

Integration with existing inspection workflows can further enhance efficiency and reduce maintenance costs.

ACKNOWLEDGMENT

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REFERENCES

- [1] J. Juneja, “Scalable Localisation and Mapping of Textured Scenes,” M.Eng. Thesis, Oxford, 2014.
- [2] J. Juneja, “Simultaneous Localisation and Mapping (SLAM) in MATLAB,” 2013.
- [3] B. Triggs et al., “Bundle Adjustment – A Modern Synthesis,” Springer, 2000.
- [4] S. Thrun et al., *Probabilistic Robotics*, MIT Press, 2005.
- [5] D. Scaramuzza, F. Fraundorfer, “Visual Odometry,” IEEE R&A Magazine, 2011.
- [6] T. Bailey, H. Durrant-Whyte, “SLAM: Part I,” IEEE R&A Magazine, 2006.
- [7] R. Mur-Artal, J. D. Tardos, “ORB-SLAM: A Versatile and Accurate Monocular SLAM System,” IEEE TRO, 2015.
- [8] J. Engel et al., “LSD-SLAM: Large-Scale Direct Monocular SLAM,” ECCV, 2014.
- [9] J. Engel et al., “Direct Sparse Odometry,” TPAMI, 2017.
- [10] R. Murphy, *Disaster Robotics*, MIT Press, 2014.
- [11] S. Tadokoro, *Rescue Robotics*, Springer, 2009.
- [12] J. Queralta et al., “Collaborative SLAM in Harsh Environments: A Review,” IEEE ICMR, 2020.
- [13] T. Qin et al., “VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator,” IEEE TRO, 2018.
- [14] C. Cadena et al., “Toward the Robust-Perception Age,” RA-L, 2016.
- [15] C. Forster et al., “SVO: Semi-Direct Visual Odometry,” ICRA, 2014.